What Is Replication and Transactional Replication

What is Replication

- SQL Server replication is a technology for copying and distributing data and database objects from one database to another and then synchronizing between databases to maintain consistency and integrity of the data.
- In most cases, replication is a process of reproducing the data at the desired targets.
- SQL Server replication is used for copying and synchronizing data continuously or it can also be scheduled to run at predetermined intervals.
- There are several different replication techniques that support a variety of data synchronization approaches; one-way; one-to-many; many-to-one; and bi-directional, and keep several datasets in sync with each other.
- Types of Replication:
    - Transactional Replication.
    - Merge Replication
    - Snapshot Replication

Transactional Replication

- SQL Server Transactional Replication is a real time, database level, high availability solution, that consists of one primary server, known as **Publisher**, that distributes all the database tables, or selected tables known as articles, to one or more secondary servers, known as **Subscribers**, that can be also used for reporting purposes.
- As the name indicates, Transactional Replication depends on a synchronization process with the SQL Server Transaction Logs.
- Components of Transactional Replications
- **Publisher :** The Publisher is a database instance that makes data available to other locations through SQL Server replication. The Publisher can have one or more publications, each defining a logically related set of objects and data to replicate.
- **Publication database** : The publisher is a database that contains a list of objects that are designated as SQL Server replication articles are known as publication database. The publisher can have one or more publications. Each publisher defines a data propagation mechanism by creating several internal replication stored procedures.
- **Publication :** is a logical collection of articles from a database. The entity allows us to define and configure article properties at the higher level so that the properties are inherited to all the articles in that group.

Transactional Replication (contd..)

- **Articles :** An article is the basic unit of SQL Server Replication. An article can consist of tables, stored procedures, and views. It is possible to scale the article, horizontally and vertically using a filter option. We can also create multiple articles on the same object with some restrictions and limitations.
- **Distributor :** The Distributor is a database that acts as a storehouse for replication specific data associated with one or more Publishers. In many cases, the distributor is a single database that acts as both the Publisher and the Distributor. In the context of SQL Server replication, this is commonly known as a “local distributor”. On the other hand, if it’s configured on a separate server, then it is known as a “remote distributor”. Each Publisher is associated with a single database known as a “distribution database” aka the “Distributor”.
- **Distribution databases :** Each Distributor must have at least one distribution database. The distribution database consists of article detail, replication meta-data and data. A Distributor can hold more than one distribution database; however, all publications defined on a single Publisher must use the same distribution database.
- **Subscriber :** A database instance that consumes SQL Server replication data from a publication is called a Subscriber. The subscriber can receive data from one or more publishers and publications. The subscriber can also pass data changes back to the publisher or republish the data to other subscribers depending on the type of the replication design and model.

Transactional Replication (contd..)

- **Subscriptions :** A subscription is a request for a copy of a publication to be delivered to a Subscriber. The subscription defines what publication data will be received, where, and when. There are two types of subscriptions
- Push subscription: Distributor directly updates the data in the Subscriber database.
- Pull subscription: the Subscriber is scheduled to check at the Distributor regularly if any new changes are available, and then updates the data in the subscription database itself.
- **Subscription databases :** A target database of a replication model is called a subscription database.
- **Replication agents :** SQL Server replication uses a pre-defined set of standalone programs and events are known as agents, to carry out the tasks associated with data.
- The **SQL Server Snapshot Agent** that prepares the starting snapshot file that contains the database objects to be replicated, you can imagine it as a full backup file
- The **Distribution Agent** is responsible for copying the initial snapshot and the cumulative logs to the subscribers
- The **Log Reader Agent** is responsible for monitoring the SQL Server transaction log in the publisher database and copy these transactions from the transaction log file of that database to the distribution database, to be copied to the subscribers by the distribution agent later, as shown below:

Transactional Replication Diagram